Ep. 371: Paul Krake on AI's Post-Cyclical World, 10% Unemployment Risks, and the End of Friction

Podcast: Macro hive

AI Executive Summary

1. Core Thesis: Paul Craig argues that technological advancements, particularly artificial intelligence (AI), containerization, and enterprise software, have fundamentally altered the global economy by reducing economic friction and business cycle volatility. He posits a shift toward a "post-cyclical" world where traditional macroeconomic models no longer hold true because corporations are becoming more efficient at production while labor's share of GDP declines. This structural shift leads to a future characterized by significant labor displacement (potentially reaching 10% unemployment in knowledge sectors), rising income inequality favoring capital over labor, and reduced efficacy of monetary policy due to supply-side rather than demand-side constraints.

2. Key Arguments & Data:

3. **Fading Business Cycle:** Economic volatility has collapsed since the 1970s as technology removed "sand from the gears" of production. Corporate margins have increased linearly (from ~1% in the mid-70s to over 14% recently) while the labor share of GDP declined, coinciding with a reduction in standard deviations of quarterly volatility.
4. **AI and Employment Displacement:** Unlike historical tech waves where new industries replaced old ones (e.g., agriculture to manufacturing), there is currently no replacement industry for displaced knowledge workers. AI targets human cognition, displacing bottom 15% of staff without viable sector alternatives like elder care or health care which cannot scale the same way.
5. **The 2030s Horizon:** Craig projects a "hockey stick" curve where job displacement accelerates starting in the 2030s. He estimates knowledge work (lawyers, accountants, social media) will displace roughly 10% of workers, potentially tripling US unemployment from ~4.5% to ~9%.
6. **Profit vs. Labor Share:** AI targets the cost side of the equation (cost reduction), not just revenue growth. While capital sees record margins

and profitability, the tax base is shifting; 80-88% of tax revenue relies on labor, which is in structural decline compared to corporate taxes.

7. **Monetary Policy Inefficacy:** Since unemployment and inflation are driven by supply-side labor shortages rather than demand failures, monetary policy tools (interest rates) will become irrelevant to fixing the issue. Wage stagnation and potential deflationary pressure could occur even as margins remain high.
8. **Asset Valuation Shifts:** The "Lucas Tree" model (Google/Microsoft generating infinite free cash flow with low CapEx) is becoming unsustainable due to AI adoption costs and demand constraints. Investors should expect multiple compression for these mega-cap firms over the long term.

9. **Investment Strategy:** Traditional volatility management and market timing are futile on a 20-30 year horizon. The S&P 500 has never lost money since 1937; therefore, long-duration equity holds without leverage or hedging should be preferred for wealth preservation.

10. Actionable Takeaways:

11. **For Investors:** Adopt a "long equity only" strategy over permanent timeframes (20-30 years). Do not hedge against volatility in the long term as it destroys compounding. Focus on diversified portfolios rather than trying to pick winners in AI or betting against bonds, as bond performance is unpredictable due to shifting economic structures.
12. **For Professionals:** Avoid industries dominated by cognitive work (law, accounting, general coding, investment banking) as these are high-risk for automation. Pivot toward "atoms" and molecules (physical work, robotics), where robots lag significantly behind AI, or fields like health care and psychology that require human judgment.
13. **For Entrepreneurs:** Be prepared to operate businesses without relying on headcount growth. VCs increasingly fund firms that deploy AI to replace staff rather than expand teams. The barrier to entry is high, requiring deep product-market fit in AI deployment rather than traditional job creation.
14. **Policy Perspective:** Consider tax models that reward employing humans (e.g., negative income tax or hiring tax breaks) rather than

relying on a shrinking labor base for government revenue and social safety nets.

Full Transcript

Welcome to the MacroHive Conversations podcast. I'm Bilal Hafiz, Head of Strategy at MacroHive. Before we dive in, a quick update. Following MacroHive's acquisition by the BGC Group, BGC has launched a new regulated business called MacroHive Markets. For the first time, onboarded clients can also access execution services via MacroHive Markets. This new workflow will allow macro-institutional investors to benefit from strategy and quant insights, alongside access to global liquidity. Please contact Andrew Simon or Gail Mozibriyan

Bloomberg or click the link in the description below for more details on macroeconomics and its service offering. Now onto this episode's guest Paul Craig. Paul is an investor, strategist and independent researcher with three decades across macroeconomics, policy and capital markets. In 2011 he founded the research platform View from the Peak after macro roles at Chorus Capital, CACS & Associates, More Capital, Goldman Sachs and Macquarie Bank. Now on to our conversation. Greetings and welcome, Paul. It's fantastic to have you on the podcast show. Well, thank you. It's good to see you again. It's been a while. It's been a while, absolutely. Now, since we last spoke, you published a book, Growth Without You, How AI Rewise Work, Wealth and Investing. So congratulations on the book. Thank you. You know what? I think you hear a lot of people complaining about the process. I actually thoroughly enjoyed it. I thoroughly enjoyed it. I had very good reasons for writing it, which we'll get into a little later.

Yeah, look, editing's a pain, but anyone who writes a newsletter or has errors in the investment game knows about editing reports, it's that on steroids. But it was a great process, and more importantly, I learned a lot during the whole thing. So it was great. That's great. And we'll talk more about that as well. Before we go into that, our audience always loves to learn something about the origin story of our guests. So it'd be great to hear in your own words what you studied at uni, was it inevitable you'd end up in finance and then research and so on, and some of your key milestones career-wise. So back, well, it's going on. four years ago. I had a business called Climate Transform, which was an interview series basically around

clean tech investing. So a very long story short. We did about 600 interviews over four years. And the one thing that came up every single time is that everyone who started the company had an origin story. But not everyone has an origin story.

But I didn't really have an origin story. So I do believe that you sort of fall into things and I enjoyed economics. I didn't start a trading business at the age of eight. I wasn't a climate guy who ran naked through the forest and felt at one with nature and all this sort of stuff. I liked it. And I try to look long story short, I try to get into law school, which is undergraduate degree in Australia. I missed out by two HSC points of getting into Melbourne University Law and went and did economics at Monash University instead. And, you know, frankly, I just enjoyed it and there was no natural. acumen or anything like that. I think it was, you know, you tend to be good at the things you like. And that's where it sort of evolved into. And then, you know, career-wise, you went on to work for Macquarie, some sort of, was it goldman's, and then use some hedge funds as well. I mean, what were the main learnings there? So it sounds like you were sell side and buy side. Sell side by side. The observations are that I don't get that job at Goldman Sachs today.

Right because I'm not bilingual. I don't have a master's degree Yeah, this is when if you had a little bit of chat you could be a salesperson at a large investment bank I think those days are behind us and and frankly I think this you know the world we have today as much more meritocracy in that regard because good people get good jobs today and Not that people who worked at Goldman back in the 90s weren't good at what they do because it wasn't very Yeah, an elite group of people which was a great pleasure to work for But the pie was smaller back then, and I wasn't competing with kids from India who go to IIT and the like. So I do believe that luck plays a big role in our career, in everyone's career. So I was lucky enough to, at the end of the Asian crisis, I got caught up in that and walked into a job at Moore Capital about two weeks later.

always wanted to live in New York and I met from Hong Kong to New York and sort of, you know, life evolves from there. And so, you know, you're lucky enough to, you know, keep from the bush, the bush in Victoria

shouldn't go and work for Lewis Bacon and Bruce Covner. And I was lucky enough to do that. And, you know, I was lucky enough to meet Dwight Anderson, who seeded my hedge fund in, you know, in 2004. And, you know, these things all sort of fall into place. And I think that, you know, through careers, you have good luck and bad luck. And I've certainly had more good luck than bad. That's great to hear. Now, you have written this book, Growth Without You, and it obviously has sort of an AI sort of focus. So what was the rationale for writing a book? I mean, many people write research notes or blogs, or they have a sub-stack, but instead you decided to... I mean, it doesn't sound like it was that painful, at least pre-editing. It wasn't as painful, but why a book? So, well, I'll give you the reasons for writing it sort of intellectually. And one, well, it evolved, frankly.

And the first reason I started writing it was my belief that the way that you and I were taught macro in that very traditional sense, right? Whether it's through an economic lens at investment banks, whether it's at a hedge fund with goins of the industry like Lewis and Bruce Kovner and these folks, that world for me doesn't work anymore. And I wanted to know, and I really wanted to find out why. And I think it was really sort of twofold. One is an observation that we just don't have recessions like we used to. And there's a bunch of reasons that people say which predominantly revolve around government policy. And funnily enough, I'm writing a report for clients this week talking about called, it is different this time, right? And you put it in the book, it's littered with that.

connotation that is different. To say that to a lot of macro people is to get laughed at because it's never different. Why is the US not had an organic business cycle recession excluding COVID, which we can agree was an organic in 17 years? Why does Australia and the Netherlands go decades without recessions? And I wanted to find out why, and I think I've got some pretty good evidence to why, and we can talk about that in a sec. And the second reason where it sort of morphed as much from a book about what I call heading to a post-cyclical world to one which looks really detailed at where we're going from an unemployment perspective is a dinner I had with my son in college. He was a freshman. He's a sophomore this year, a freshman last year, and I took a bunch of his mates out for dinner.

And he's at Northeastern University in Boston, very, very good school. And one of his mates is a third year computer science major. And he said to me, Mr. Craig, when I came to Northeastern, sort of paraphrasing the world was my oyster. And today I think I'm unemployable. And Blar, when you hear that I'm a 20 year old, that's a kick in the guts. And so I really wanted to look at whether or not this assumption I was making of, yeah. 2035 we get to 10% unemployment. There is a method in the madness of how I get to that number. Whether that was actually factual. And so that sort of drove the second half of the book. And there's a reason it should have stopped at 300 pages and ended up being 500 because I didn't know where to stop. And so I did a deep, deep dive into that. But the real premise of the book is about why this is different and why things, the way we learn them don't work like they used to. What's the implication for monetary policy, fiscal policy for asset allocation. Is there a role for bonds in portfolios? Is there a role for volatility management in portfolios? All these sort of things all come up and that all stems from the fundamental changes that I've seen in the economy over the course of the last 15 years. Okay. I mean, perhaps you can start on the business cycle side.

in your earlier parts of the book, you talk about the business cycle is fading, and it's not just an AI story. You talk about numerous technological shocks from containerization to price software, and obviously globalization and so on. Talk me through how that removes or fades, reduces the business cycle. Sure. Again, going back to the point about changing the way I think about macro. I've changed my thinking from macro being a top-down story. where people assume laws of nature effectively to a bottom-up story, which is how do the underlying constituents of the economy, consumers, companies, government, behave? So if you think that any economy is lots and lots of consumers, many, many companies, and we'll concede too much government, right? How does that behave? How does that behave? And how does technology influence those things? So You go back to the 1970s and it took you 90 days to get a check to get paid for services. You sent a paper invoice, got a paper check back, and that added clear. It took weeks to clear at the bank and the like. You did your sales forecast based on the quarterly conference cycle. You had no price transparency and you produced locally and you sold locally and the like. Over time as Containerization is a great example of this. The standardization of container

ships allowed you to send things cheaply overseas, not only send it to sell overseas, but to actually start producing overseas as well. Containerization, globalization, enterprise software is the big one because that's the closest thing we have to AI in terms of that transition. What we've done in the last 50 years is remove companies, have removed sand in the gears so what I call the reduction of economic friction and whether it's things like inventory to sales ratios whether it's you know things like you know any really any kpi you want to look at about efficiencies of companies right its companies have just gotten more efficient and the best example is to look at corporate margins.

So corporate margins in the mid-70s were 1% for the S&P, and the last reading they had in Q1 was 14 points set. Now, we can debate whether that's anomaly or right, but what's clear is that they've gone up in a straight line. And what that coincides with is that it coincides almost tick for tick with the decline in the labor share of GDP. So whether it's Blanchard and Simon's talking about the great moderation, the theory behind the great moderation, they didn't actually coined the phrase originally, whether it's about the great moderation, whether it's about reduced economic volatility. So the standard deviation of quarterly volus collapsed around the world, whether it's, as I said, inventory to sales, whether it's headcount, revenue per employee, all these numbers move in the same direction. And it all happened as capital has become more influential and labor has become less important. And what does AI do? AI is the final layer.

a friction removal for many companies around the world, because what you do is you take the most expensive input for companies, which is human cognition, and you're replacing that with cheaper, eventually replacing it with cheaper artificial cognition. Again, it's not a linear pathway. We're very early innings, all this stuff that we know, but that's the basic premise of all of this. economic volatility for companies as they become more efficient the boom bus cycle goes goes away there's less corporate bankruptcies there's margins don't mean revert so Jeremy Grantham's claim that there's something wrong with the global economy if margins don't mean revert well Jeremy I'm sorry you're wrong on that one because margins haven't been mean reverting and it's all because of technology deployment making companies just better and better and better and better at producing stuff.

And if you aggregate all that, that's where you get the diminished role of the business cycle. Yeah, I guess a couple of questions follow on from that. One is the decline of the business cycle. I mean, is that to do with economies moving towards becoming a service-based economy and away from manufacturing? 100%. It's all part of the mix. So that's definitely part of it. It's a major, major part of it. But we've got better at producing goods and services as well. So just think about CGI and movies as a good example. Right, so I can't use the Odyssey because Christopher Nolan doesn't use CGI. But Moana or something or the upcoming Spider-Man. Spider-Man, whatever it is, right? You can make movies much more efficient, right? And so, well, again, I'll give you a really good example. It's encyclopedia Britannica, right? Now, the Craig family was not a rich family by any stretch below when we lived in country Victoria, but Mum and Dad forked out 800 bucks in 1982 for its full set of leather-bound encyclopedia Britannica.

My dad would not let me go near that thing if I was eating or anything. This was the prize possession in the Craig household. And the Cyclopedia Britannica back in the 80s was a 250 million global business, right? Good business. Got to destroy the Wikipedia. And so, and then we can talk a little bit about sort of the growth connotations of that. So actually, innovation in this scenario is actually bad for growth because Wikipedia is zero contribution to GDP growth. But you've lost a 250 million business in the form of of encyclopedia Britannica, Spotify versus CDs, all these sorts of things. And these are all just the economy being more efficient on the services side, not just the good side. And so it's certainly part of that mix. But again, it's all in the mix. And the other question, and this is something I often think about, is you mentioned the corporate profit share of GDP has gone up and labor share has gone down.

At the same time, the unemployment rate has also been trending down or to remain low for a long period of time. So how is it that companies almost seem like they don't need as many people, but at the same time, at least according to labor stats, you know, we're at full employment. Yeah, and then so I think that the lot of that has to do with you know, as in you know, under employment in multiple jobs because how holding multiple jobs has never been more prevalent. You know gig economy work is very common.

So very look at the average Uber driver doesn't just do Uber. They do other things. So it's as much about income supplementarity as it is about a singular career. And in the book, it's the one of the big challenges about AI is the, you know, the question of, well, we've always replaced jobs with other jobs.

Why won't this be different? And I spend a lot of time sort of dispelling that. And I do think that at its core, AI does challenge the last of the fundamental monopolies that people have, which is cognition. And so when you do that, what other careers can pop up to replace that at scale? And the other thing just to square that circle is that it's very important when you think about that premise is that every technology that is being disrupted. right, whether it's agriculture and manufacturing. Manufacturing didn't disrupt agriculture, right. Manufacturing was booming from 1910 to 1920 as the models he took off, and agriculture in the United States was roughly flat. It was down to agriculture employment was down about 5% during that period. All the employment came from immigration. People didn't go from farms to factories, they went from ships to factories, right. And so the trouble is that every So when agriculture went into decline in the 1920s, you already had booming manufacturing. And when China took away manufacturing jobs in the 70s, 80s and 90s, when you had outsourcing in the 70s, 80s and 90s, services were already booming and knowledge work was booming. You don't have that replacement industry on deck today. And that for me creates the a real structural problem going forward when this technology gets good enough to start replacing people. Yeah. I was about to say on mass, but accelerate the replacement of people. And in your book, I mean, you talk about there's a potential path to 10% unemployment. So that's quite a dramatic forecast there. And that's related to what you just said, presumably. Yeah. And the math is really straightforward. JP Morgan looked at this last year. And there's an area required a recession where I looked at this and I don't think you do.

My pathway just takes a lot longer. So knowledge work in the US, you and I are knowledge workers, lawyers, accountants, social media people, that is 45% of you of the US economy. You displace 10% of those workers and they don't get into going to other things. So obviously, this is a long cycle. Well, that'll take you from 4.5% roughly unemployment US to 9%. So

you're nearly there. And that's before robotics that's before other parts of the economy that can be influenced by AI. Look, can we see a world where we have 10% less lawyers, 10% less accountants, 10% less social media people? Yeah. The question is how quickly, if at all, they find something else to do, and that's where I think it becomes very challenging, particularly going into the 2030s when this technology is not just aiding you and I and everyone else who's using the tools today, but actually actively displacing the bottom 10, the bottom 15% of staff at each and every firm. You know, I guess one question that comes up is, you know, whenever we've had these technological waves, there's always this fear that labor will be left behind, but invariably some new part of the economy opens up and that employs people. Isn't there a chance this could happen as well where AI just opens up a whole way for new jobs that we haven't really been able to foresee today? And so it'll just be humans running, you know, lots of different AI agents and whatever that new job is, they do that. Sure. So three things to that. One is every display sector in the past, the new sector was already in place, right? And we certainly don't have that today.

Right. Number two, the area that gets talked a lot about is things like health care and, you know, elder care and health care providers and like. That software developer at 120,000 a year that gets displaced by code can go into elder care at 45,000, and there are real economic consequences to that. So we're not at the moment. A, we don't have the sectors to replace it. And B, the suggestions. I've heard life coaches, I've heard elder care, all this sort of stuff. These are not like for like jobs. And so I think that that's the second part of it. But I'll give you the third, which is something that gets talked about a lot is that it's a golden age to start a business and you speak to any VC and they're littered with unbelievably good pitch decks that come in every day for new business. My neighbour here is a 28-year-old who just got a quarter million bucks from a seed fund for an AI business around monitoring safety in factories and warehouses.

He showed me his deck before he sent that in. Do you know how many times in his deck he mentions how many people he's going to hire? Zero. Because he knows, and everyone who sends a deck to a VC knows, that they don't want to hear that you're doing anything but deploying AI tools instead of hiring people. And of the thousands of pitch decks that get sent

out every quarter around the world, very, very few of them talk about hiring a thousand people. And so again, it gets back to this notion of cognition. We are deploying cognition as a service. That artificial cognition is cheaper and frankly declining in cost versus the constantly accelerating cost of human cognition. And eventually, you'll target the bottom 5% of workers, 10% of workers, and then we get into the mid-2030s and you're dealing with 10% unemployment and all the consequences of that.

And in terms of the cognition replacement, at the moment, AI can't do everything a human can do. There are aspects of what we do as humans that AI can't quite replicate. So we know it's very good at writing code, summarization, those sorts of things, but it can't do everything. So is part of your thesis that AI will broaden and improve so that we reach ADI or these more general intelligence? Or is it sufficient that we have incremental improvement on the existing technology. So Caterpillar in 1990 had about 300 people in its finance department in the United States. That's roughly that. Today it's got about 110. And that's because of enterprise software. One of the things I believe that owning the customer is going to remain a very fundamental thing for businesses. So that's why Salesforce, ServiceNow, these enterprise software companies I think are really interesting because they're going to be the vehicle in which these AI tools will be deployed for many, many companies. So can Caterpillar take that number down by 20% through the adoption of new AI tools?

take what they've done on enterprise software that little bit further. And so at the moment, I think we're seeing AI adoption come through, not necessarily in firing, but more about a lack of hiring. Well, it's the rise of the 55-year-old man. I can say that with confidence because I was bearish on 55-year-old men as a 55-year-old man because we couldn't keep up with 30 year olds, the energy of a 30 year old and everything, they're cheaper, they're more tech savvy the whole thing. AI's completely changed this. So we're suddenly super productive. So if we're in a team of 20 and someone leaves, we're going to do what we're doing with a team of 19, which is probably being more productive than we were a year ago anyway.

All right. So you just start to see this gradual attrition which starts with the non rehiring of people and the jolt stator is pretty, you know, it's bounced

recently, but it was back in 2008 lows only a year ago, right, in terms of hiring intentions. So, you know, I think that you don't have to get to AGI for, again, the bottom 5% of work is to be under threat, just to be not replaced. And so, and again, These tools are not there today, right? So if you and I want to write a substack post, we could probably, you know, we could stick one of our old reports in a chat GVT or Claude and get a 200 word substack post and shoot that easy. That can be done. Code can be written. We know this sort of stuff. You know, things like radiology and these sorts of tools are really awesome at the moment, but they're not replacing a human radiologist yet.

But again, it's a gradual adoption and this is why I think the just one last point the issues of the bears saying well there's no ROI in AI at the moment. I completely missed the point because we're in the trial and error phase. There are so many AI guys, companies having so much fun right now because they're getting paid to burn tokens, which is exactly what people did in. from 1996 through 2004 when you adopted the internet. How did banks know that online banking was going to be a thing? It was trial and error. How did airlines know the majority of tickets would be sold online by 2005, trial and error? And if you look at the work of Brevolskan and these sorts of folks, they have done, again, the adoption of these tools is all about embedding them into the processes of companies.

And that takes time. And so the fact there's no ROI in many sectors in AI right now doesn't mean that people aren't going to keep trying to find. And just putting kind of my sort of Keynesian economies hat on, if we do see this rising unemployment and workers aren't earning as much as before, where's the demand going to come for all of these companies? So you and I as writers of this stuff, we have the luxury to look 10 years forward, which investors don't have. So let's put the world up into two five-year blocks. This five years is golden age for equity investing because you've got the growth that you know I eventually as a growth killer and we can talk a little bit about that in a minute but you know but the growth elements data data center rollouts massive increase in electricity production all that the the infrastructure around that right which is great for GDP that's all now that's all they know and they're all the negativity around you know cap expand and all this sort of stuff it's getting spent that's great for Right? And so that's

now the unemployment side of things. We're not taking up yet at the moment because these are enhancing current employers and not seeing the loss of jobs.

Look at that. So when Palantir makes a claim that they can do 10x the revenue with 10% less staff, that's really looking into the 2030s that they eventually get to that. And there's a non-linear pathway where we slow the start and accelerate into a hockey stick curve later on. But for the next five years, you've got adoption by tools. You've got eventual ROI. You've got Microsoft and eventually Open AI and Anthropics selling a lot of the product around this sort of stuff, Salesforce selling a lot of product. More to the point, you go from the providers of AI services being dominant to the beneficiaries, which is a couple of hundred companies globally, which dominate the FT every day, to tens of thousands of public and private companies that could add 25, 50, 100 basis points a year every year to margin because they start adopting tools and they get rid of.

20% of HR and 20% of finance and 20% of logistics and all this sort of stuff and replace them with these tools. And so the next five years, let's back at 2030, boom time. After that, that's where the technology starts to get better. That's where the bottlenecks such as electricity that can slow the adoption of AI. And let's be clear, bottlenecks are good for jobs because the slower these tools get adopted, the slower the job displacement, right? as you get more electricity available and the data centers are there and you're adopting larger scales and more people start to get fired. The next two or three years, you might lose two to three tenths a year in terms of AI, but then it goes to four tenths, five tenths, six tenths, seven tenths, and suddenly we wake up and there's a million jobs a year being displaced because of AI. That's the hockey stick approach. That's classic tech adoption. We've seen that sort of that hockey stick regularly.

When you get into the 2030s, that's when you start to have real problems in terms of the consumption side of the equation, right? Because let's face it, GDP is how many workers you've got, how productive they are, and how many hours they work, right? Now, you can have them be much more productive, but having many fewer workers, right? And so that well could be a wash. It's very clear that AI tools will predominantly target the cost

side of the equation. So they're not really drivers of revenue as much as they are drivers of cost. And if innovation drives cost, it doesn't appear in productivity statistics, right? So this is Robert Solo saying, you know, computerization is everywhere except in the productivity numbers, right? Because why was that the case? Because it was targeting cost and not revenue.

And so you have a lot of these sorts of things kicking in where, you know, at the end of the day, people don't have jobs. You've got 10% unemployment, you've got a big consumption hit. Undeniably so. You've got a real fiscal problem. And the fiscal problem is less on the spending side of things and more on the revenue side of things because, you know, you've got 80, 88% of the tax base is funded by labor. Labor is in structural decline. 9% of the tax base is corporate taxes. And what we know about corporates, they're flexible, they're mobile, they're litigious, and they will fight tooth and nail to stop paying their, what I would argue is their fair share of tax revenue when labor is in structural decline. So capital needs to be paying more tax going forward and labor needs to be paying less. And so that gets to a stage where you've got some real problems in the 2030s because I can see Europe finding ways to solve these problems because Europe tends to find ways to solve these sorts of problems. Japan probably finds a way to change. Japan's a big winner in all of this because you want to shrink labor force in the era of AI, right? And so, but the US, I don't know if the US political system that has done nothing regarding social media regulation. We can't stop teenage kids from hurting themselves, pass legislation about that. How do we pass tools about artificial general intelligence?

That's an interesting take because most people kind of view the US as being the big winner in the whole AI story. But from what you're saying, at least in the long run, actually, the US may end up losing in some ways in terms of adapting society for the technology. Well, think about how we define winners and losers, right? So capital win, labor loses, right? So 50% of listed equities are owned by the top 1%. Happy days for them. Right because capital continues to win margins continue to rise profits continue to grow up until the point of the time you got a revenue problem with demand but you capital wins right but if you are Labor loses all over the world that

particularly loses in the US where you've got income inequality already at very very high levels and Instead of the bottom 50% being victims of income inequality like it is today, this is the bottom 80% because job displacement is targeting the suburbs of Chicago and Atlanta and Greenwich, Connecticut and places like that. By the way, this is a global thing. But the US is not being good historically at social safety nets. And so if people can't get jobs, Whether it's the Korean approach of robot taxes, my best solution is the only solution I can really think of is you actually are providing tax breaks for employing people. So instead of paying FICA to fund social security, you actually get a tax break or Milton Friedman talked about back in the day.

talked about universal basic income in his own way, calling it a negative income tax, similar sort of thing. But the US is not good at passing legislation. Look, US tech companies are the world's most profitable and largest behemoths because they have no regulation. This is the most under-regulated industry in the history of mankind. There is nothing that holds back Microsoft on a daily basis. There is nothing that holds back Google on a daily basis, right? And so, you know, US government historically not being good at this Republican party generally not good at thinking that the market can't solve markets will solve all problems. Markets won't solve this problem because the market because if a company is rational, they'll replace expensive human cognition with cheap artificial cognition. That's the rational thing to do.

Right. And again, there's whole social connotation too. And in terms of the AI trade, you know, we've gone through a few different versions of this versus hyper scalars than everyone has been in that sort of AI infrastructure trade. I mean, what's bigger picture? What do you think the right trade is for this? So one of the sections in the book talks about how the models that we've used for portfolio management, like 60-40 portfolios, volatility management, all those sorts of things just don't work. Just do not work. If your goal is maximizing return and you have a long duration time frame because the number thing that annoys me most about this industry is that we being the market and the punditry community rarely gives a time frame when they say they're bullish or bearish.

So you could be bullish on bonds for the next three weeks and bearish on bonds for the next five years. So saying you're bullish or bearish is irrelevant if you don't have a timeframe attached to it. And so what do we know? We know that on a 20 year rolling basis, if this is your long duration, if you're looking, put your kids through college, generational security, those sorts of things, on a 20 year rolling basis, the S&P has never lost money since 1937. Why would you ever hedge in that work? If you're just putting money away for retirement and the like, trying to predict the next 5% down or buying puts in the S&P just doesn't work. On a 30-year view, investing 100,000 at 10% versus 100,000 at 8%, which could be hedging cost, fees, friction, whatever. It's a 41% difference in terminal wealth.

over a 30 year period, right? And so I looked at a lot of this through, you know, this notion of efficiency going forward and the like. And if markets are going to be more efficient and the economy is less cyclical and margins don't mean reverse, you just want to be long equities, no leverage, right? And just be holding diversified portfolios, you know, through everything because, you know, how many people lost their shirts trying to hedge the Iran war. And the Iran war in different guises has happened every year since the dawn of time. And what happens is you sell into the hole, you don't catch the rebound, and you just lose compounding every time you try to hedge. So the AI trade for me as I slowly turn into Jack Vogel is just to stay long. If I'm talking to a guy next door, I'm just saying, buy an S&P portfolio, stay long, don't try to pick winners and losers, don't try to market time and just do it. Now, I get that most of the managers that we speak to do not have that luxury because they have to volatility hedge, they have to manage vol, et cetera. I get that. Their world is obviously suboptimal versus those worlds of people with permanent capital. The AI trade's got some issues right now.

there's a guy, probably someone you know at the University of Chicago called Robert Lucas. Robert Lucas in 1978 talked about the Lucas Tree. The Lucas Tree was a company that basically could just churn cashflow, churn free cashflow without reinvesting. Basically, you plant a tree and every year you pick the fruit and it just grows back and you maybe water it into a little bit of maintenance, but you've done all your capex in the past. Google and search is the ultimate lucas tree. If you look at the enterprise to CAPEX

ratio that went from about 8% in 1980 down to 1% in 2020, that's because companies grew their market cap without spending money on CAPEX. Those companies deserve stratospheric multitudes. Suddenly, you do the AI spent and you start, in Google's case, you go into negative free cash flow.

Well, if you've got negative free cash flow, you're not a Lucas Tree business anymore, right? You're an infrastructure business. An infrastructure business don't deserve software multiples. And so as this Lucas Tree model gets tossed to one side by Microsoft, by Google, by Metta, etc., less so Amazon, is that the multiple for those companies, even though profits are strong, the multiple must come down, right? And look, There are booming profits. You can argue whether there's a circular economy thing going on, which there is. We can debate that. But at the end of the day, the multiple, and we've seen it all year, the multiple for these companies must come down. There's a debt dynamic around some of the less strong AI providers, Oracle Coreweed. Anyone who worries about Google or Microsoft from a debt perspective needs to look at a balance, because they're no problem.

Right. You know, CoreWeave is a fragile business. And Oracle has become a fragile business. And so, you know, if we were to get some defaults on these sorts of companies, I wouldn't be shocked. But it doesn't mean that the innovation, which I think that AI is going to be second only to, you know, the great innovations of 18, 79, and 40 electricity, sanitation, you know, these sorts of refrigeration, these sorts of things, combustion engines. AI is the 21st century equivalent of those and eventually it will be completely transformative, but it doesn't mean you don't have boom bus cycles in the middle of them I think there'll be less of it in this case, but you'll go the trade was easier three years ago than it is today Understood and in terms of just want to round off with a final question from a policy perspective then what's implication here monetary policy becomes irrelevant Because there's nothing that the Fed can do to fix a supply site problem And so the Fed's role becomes, in a world which I describe as going post-cyclical, as in cycles basically are greatly reduced, the Fed's role becomes a policeman of financial regulation, like so deal with an SBB when it pops up and the like. And frankly, since 08, the Fed has proven to be pretty done good at that, right? It's done a pretty good job at that. But if you can see in an economy where AI targets the biggest expense for pretty much every company, which

is people. There's no wage inflation in that world, right? There's no inflation. Yeah, yes, we might have copper might go to 20,000 a ton, sure, but if wages are stagnating, there's no inflation in that world, right? And so wages down, unemployment up, the Pavlovian response to the Fed is cut rates, right?

Pavlovian response to the ECB with falling inflation cut rates. And yet you could be cutting rates into record margins, record profitability, booming stock market, all stemming from the fact that this is a supply side issue that monetary policy can't fix. And so the Fed loses its efficacy in terms of that. And we talked a little bit about the revenue side of taxation before, but I've never been a bear on U.S. fiscal sustainability. I thought it was a very open-blood narrative. I've got some concerns going into the 2030s about this because of the revenue side predominant. Yeah, they spend too much, but they're obviously going to have to spend more in terms of social safety nets as unemployment right and organically rises.

Okay, that's great. Well, we've talked a lot about the book and I urge everyone to buy the book, Growth Without You. I did want to follow up with something you said about, I think it was friends of your son and I'm not sure if your son is himself, into the job market for youngsters. So, you know, if there is some youngster leaving university this year, you know, we're recording this in the summer, so people will be graduating soon, depending which hemisphere they're in. What advice would you give them? So, I'd say the following. It's really tough because You and I being good at AI hurts kids coming out of college because you and I don't leave jobs as early and the like, and it's not that natural life cycle of employment within companies. You don't have to look at the legal profession. Law graduate hiring is collapsing. And that's a function of co-counseling and Harvey being these vertical governance tools that are being adopted at major law firms around the world. And frankly, by hedge funds and VC firms, they're all adopting these tools.

You want to be in the atoms and molecules business, not the bits business, right? Law recounting, consulting, even investment banking to a lesser extent. These are tough businesses in the world of AI. It's a really tough business in the world of AI. We did an hour to talk about the implications of

what this means to lead with the profession and the like. So you want to be doing you want to be in the make stuff business. There's the stuff that's enhanced by by AI. Secondly is that robotics is decades behind AI, behind traditional AI. Robotics is hard, right? You know, I haven't met the most most forms of cancer will probably be cured before a robot turns up to your house to fix your sink.

Right, because it's easier to train AI versus train robots because it's not just, you have robots in scenarios and you just don't do that with data. You need physical scenarios to do a lot of that. So robotics is much harder. And so, but again, the cliche is about, you know, use AB, be proficient in AI. Yeah, everyone has to be that. But the whole, I think there's one thing to really be careful of, not everyone can start a business, right? Then this notion of, you know, go and start a business, not that easy. When over 45% of US venture capital money goes to Harvard or Stanford graduates, it's not that easy to start a business at scale. My son's doing psychology, and psychology I think is going to be one of those things which will be really enhanced by AI. Medicine, there's nothing negative in terms of the medical profession around AI. It's amazing, just amazing. That's a great thing.

you know, doing law or accounting, you know, these professions are going to be tough. Coding is going to be tough because you are competing with something cheaper and eventually approaching the equivalent of what you can do as a human. That's great. Now, you know, you've obviously written a book. Well, there's some books that influence you. I've got that. I did my list because I was actually, I actually did my list and actually asked my wife about it was as well. So on the nonfiction side of things. The book that is littered through my book and it's the reason one of the academic sort of foundation of it is a book called The Rise and Fall of American Growth by Robert Gordon at Northwestern University. Robert Gordon hits 900 pages and he will talk about how the standardization of nails led to a construction boom in Chicago. That's the level of detail he goes into, and it's amazing. If you're nerdy and into this stuff, it's great. But he talks about every great innovation that happened through from 1870 onwards. Guaranteed reading. Thinking fast and slow is a must read for everyone who's hasn't read it by Daniel Kahneman. Two books by a guy called Carl Benedict Frey called The Technology Trap and How Progress Ends, talking about about

technology adoption in countries and how that works. Anything by Robert Caro wrote the greatest biography ever called The Power Broker by Robert Moses. Anything by David Brooks, anything by Arthur Brooks, everything by Adam Tewes, anything by Christopher Hitchin. But I probably read as much non-fiction as I do non-fiction now. And so, whether it's books like Demon Copperhead, The Amazing Adventures of Cavalier and Clay, anything by Peter Carey, The Great Australian Order Peter Carey, The Cell Out by Paul Beatty, Warfall trilogy, you know, all that stuff. What I would just quickly recommend is, you know, we get go back and buy books from 20 years ago. You know, books that won the Book of Prize back in 2005, right? These are great, great books and you don't have to be buying books that have just been released because there's hundreds of thousands, millions of books that released, got released 20 years ago, which has just stayed the test of time. That's fantastic. And so finally, where can people get your book and how can people follow your work more generally?

Mate, so Amazon is the best place to buy it. I'm trying to get it into daunting Malibu, which is if anyone knows London, that's the dream for any author. Amazon is the best place at the moment. Also, just follow me on Substack. I put out stuff on Substack pretty regularly and the like, and that's the best place because I don't think people use websites anymore. I think that's fair. I'll make sure to include that in the show notes as well. Once again, congratulations on the book and keep up the good work, Paul. It's fantastic speaking. And by the way, congrats to you. This is a staple for everybody. You're doing great work and I appreciate you a lot. No, great. Thanks for that, Paul. Thanks, Bill Hart.

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